

Active Inference Stream #009.2: "The Projective Consciousness Model and Phenomenal Selfhood" (2018)

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hello everyone welcome to the active inference live stream it is ACM stream 9.2 and it is December 1st 2020 so happy December to everyone today we are really excited to have David Um one of the authors of the paper we're going to be discussing so it will be just a great discussion today really looking forward to it welcome to team everyone team is an experiment in online team communication learning and practice related to active inference you can find us on our website at Twitter via email at our YouTube channel or our public keybase team and keybase username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work also all perspectives and backgrounds are welcome here and as far as video etiquette for live stream go mute if there's noise in your background raise your hands so that we can hear from everybody in the conversation and let's use respectful speech Behavior just some quick updates on upcoming ACM streams today we are in 9.2 December 1st talking about this projective Consciousness model paper and in the rest of December we'll be talking about two papers on December 8th and 15th will be in ACM 10 on a variational approach to scripts and on December December 22nd and 29th will be in ACM 11 on sophisticated effective inference so two cool and very interesting papers that we'll be getting to but today in act M9 we're still continuing our discussion of projective Consciousness model first we're going to have some introductions and warm-ups then we'll go through 9.2 as with previous times the projective Consciousness model paper by Will aord at all 2018 and in last conversation in 9.1 1 and 9.0 we talked about some of the groundwork and the aims and claims of the paper as well as going through the abstract and the road map today especially because we're lucky enough to have an author on board we're going to be talking about a few deeper and auxiliary questions as well as going through a few key quotations and talking about maybe some of the implications for the paper and then as always we'll return to the figures and ask what each figure is representing or invoking so cool to have this discussion

today and then different papers in 10 and 11 and then we'll have a whole new suite of papers in 2021 so get in touch with us if you want to participate first though with our nice small group today let's introduce ourselves we'll go around and give a short introduction or a check-in and then just pass it to somebody who has not spoken yet so I'm Daniel I'm in California and I will pass it to Adam uh hi um I'm Adam uh Adam saffron I'm in uh Bloomington Indiana at Indiana University as a post-doctoral researcher and um I'm very interested in Consciousness and uh recently came up with a theory I called integrated World modeling Theory which has uh a lot of overlap with uh David's work and which I find to be really fascinating and inspiring so glad to be here cool let's go to David hello there I'm David rud I'm associate professor at the University of Geneva Switzerland uh and I'm one of the coor of this paper and somebody work on trying to model the role of Consciousness including it phenomenal aspects uh in let's put it that way generally speaking biological cybernetics and more particularly active influence thank you for having me here today cool let's go to Sasha welcome Sasha hi everybody um my name is Sasha I'm in Davis California and um I study developmental neuroscience and how um synapses form in the developing brain and um really interested to be part of this conversation today cool then we'll go Alex and Stephen hi everybody I'm Alex I'm in Moscow Russia and I'm a researcher in systems management school and I pass it to Steven hello I'm Stephen I'm based in Toronto and I'm doing a practice based PhD in to uh particip atory theater approaches and the use of Landscapes to explore themes and back to Daniel I believe that introduces our nice group today as far as the warm-up questions go we'll just have people raise their hand whenever they feel like it and I'll just drop out the first two questions at the beginning and they are what interest you about studying Consciousness and what is a way that we can study it so both David and Adam I know you've thought a lot about this maybe one of you two would want to go first um here yeah David just how how did you get to this uh area it's a it's just something it be a long story but to make it short uh let's put it that way it's really related to philosophical and existential question and this the Quest for turning this type of of question into something scientific um but actually I was not that interested in Psychology when I was younger I wanted to be either an astrophysicist or a conductor you know of Orchestra but I did not interesting it's a different type of coordination that you're achieving now uh Stephen and then anyone else who wants to raise their hand yeah I'm I'm interested in conscious it's in a way it's kind of a late arrival because I um I always thought that was something esoteric that I couldn't really participate in but um and uh I've suddenly seen this has start to come Center Stage um and in both this active inference world but also being talked

about more in in in in the participatory theater environment so you know what what does it mean to be consciously aware to have imagination and what I think is quite interesting is to try and move Beyond this kind of pan psychic um approach that seems to permeate um a lot of people who actually go there um either it's either something that's ignored and pushed to the side or it's everywhere and everything in this pan psychic and what I am really interested is like how can it be engaged in a kind of in terms of how we are reflective or reflexive or participatory and what Consciousness means in terms of talking to people who aren't the same as us interesting definitely the coming from a different perspective is something that's baked into this type of a framing because yeah everyone's coming from a different perspective that's our starting point that's not our conclusion here and then there's questions about how we're going to coordinate amidst that so we can either have anyone else raise their hand on these questions or I'll just put up this special point2 question which is what is something that you have thought about differently after reading the paper or participating in last week's discussion and for me it was actually uh biking in the morning and seeing the moon just look massive and thinking about that illusion in the context of the projection and what we talked about last week as well as some other literature by these authors so I just thought what might be some differentiable experiences some little tests that we could do that would be able to capture differences between the projective model and other models of Consciousness I didn't really have an answer but it was just something I was thinking about after our discussion anyone else um after reading the paper or participating in the discussion Stephen well one thing that it was really helpful this unpacking the diagrams last week and I noticed that there was that box for the head on of of of the perspective the Consciousness perspectival Consciousness and it suddenly made me realize that yes this can Bridge more back into the embodied work because it I had a discussion about this paper before with someone who's in theater and interested in kind of some of this more kind of esoterical kind of ways of knowing but I'm trying to find a way to bridge it back to the embodied sort of inactive work and I sort of by seeing it in the in the kind of head I suddenly realize okay it still does come back to the physicality so that was really really useful for me cool cool Sasha um yeah after last week I just began seeing everything as um you know more more geometrical and thinking about um uh people and their behavior as projections um kind of yeah um recognizing that we're always seeing just one side or one perspective of someone's behavior and um specifically that that GB logo it just uh that's all I could see right with light from the different side showing the different letters GB yeah and we'll come back to GB a little bit later in this discussion any other thoughts from the panelists

on these warm-up questions otherwise I'm sure there'll be a lot to get into a lot of tangents to go off on to cool all right David go ahead no I would just I just wanted to say that there would be a lot to say about the points and questions that were raised but perhaps we can not do that now and go through the the the stream and and see your questions come about great so today in this discussion as with our last two weeks we've been talking about the projective Consciousness model and phenomenal selfhood which is a paper in Frontiers and psychology December 2018 by Will aord at all and thanks David again for coming out to the stream it's always awesome to have the authors the aims and claims of the paper we did go through in a previous week but David just I'm curious here uh how do you present this work or how do you introduce this topic to different audiences who might have familiarity with different topics like how do you frame the aims and the claims of the paper so lately I've been particularly talking in context where you know people come from artificial intelligence robotics virtual reality um with you know of course some more broader philosophical interest and I generally try to frame it starting bit like you did I mean from the question of you know what we could call biological cybernetics and you elaborated on that term before uh and how active inference for situated autonomous system you know with survival challenges is an interesting paring to understand cognition within biological cybernetics and then I come to the question of what would be then the the role in that process of what you know we in it as being Consciousness what we can say about it emphasizing the the big and important challenge of connecting let's put it that way for short you know the phenomenology of subjective experience is some aspect of it with for lack of a better word functions uh why do we perceive and imagine the world the way and others and ourself in the world the way we seem to do at a certain level and uh how that actually how can we make sense of that to understand computational in some sense process that are key to you know adapt adaptation uh you know resilience in the context of biological cybernetics cool thanks for that and I think it was really clear how you laid out that connection in the paper basically there's the functions of Consciousness which we'll go through soon and then there's the uh subjective aspects which were captured in the invariance and the PCM is like a bridge that's going to be connecting the subjective invariance of Consciousness to some of these functional attributes and so we're kind of going from point A to point B uh across a specific River and so if we can really be clear about which sites on the riverbank we're building from and then be really clear about what the bridge is and isn't then we'll have a really good discussion about Consciousness as opposed to just swirling around and thinking it does everything and nothing and it's everything and nothing at once and all these things so any

thoughts on the aims and claims from anyone else otherwise we'll just go through to the next section cool yep we have a lot of questions coming up so in uh previous weeks we looked through the abstract sentence by sentence just to understand how the authors were taking the free energy minimizing approach of active inference biological cybernetics and then connecting this to a few other areas such as geometry and the embedded psychology and uh that was the abstract in the road map we had the sort of outline of where they go they start with an introduction and go straight to the methodology and then talk about the two river banks in that previous metaphor that we want to connect the phenomenological invariance of experience and then the functional features of Consciousness and then sets up the problem in a way that hopefully could be solved so maybe uh David I'm curious like in other papers how have you structured it differently or this was in 2018 what have you been um like wishing that could have been explained more or where have have you bolstered the theory since then all right that's a huge uh question um so number one kenet Williford the first author of this paper we expected him to join I don't know if he will join later is a friend of mine and I admired for the quality of his mind is a philosopher and so the structure of this particular paper we look at today is I think deeply informed by his you know philosophical methods uh if you were to read the first paper we published on the topics in the Journal of biological theoretical biology sorry uh in 2017 you would find a two wordy though but very different way of presenting but they are totally consistent with each other and uh in both case you know it's still work in progress so there are still a lot of question question and of course we are not claiming that we explain everything by far we're trying to restrict ourself to things we think we can frame in a classical scientific framework and you know I've been working for years now in my lab and we are actually finishing a paper on that to integrate in this in the general based on the general principle of this model in addition to um visual illusion as we did recently in the Moon Illusion paper that was published in 2020 uh mechanism of effective and social appraisal and processing and social perspective taking based on projective geometry notably but not only um to model and understand how artificial agent virtual or robotics as a model of ourself if you wish can do theory of mind and how that plays role in Emotion regulation uh adaptation of behavior um and that's basically what I've been doing and in interestingly without taking too long there are some studies from psychophysics that that show or that you know based on empirical data that if you are confronted with an object or something you you you feel is dangerous like inance a snake or for some phobic for instance uh the intensity of your fear will be a a function of the inverse more or less the inverse of the distance of the object and that has

been formalized as being you know something you can identify with an empirical psychophysical law called Stevens Law with a certain type of exponent and as it turns out actually the projective geometrical framing just give you that naturally it transform information notably the size the apparent size of object according under certain constraint to AO in that is inverse from the distance more or less so you can recover things that have been documented before and understand how they may play a role in appraisal something I feel when it's closer it's bigger it takes more of my field of Consciousness if you wish to make it simple and so one way of motivating my behavior of for instance avoidance is imagining myself farther so the thing is smaller takes less space and as a result it has less impact following the very same law we were discussing and therefore that will minimize my free energy and I will move away and I do that with robots and things like that huh very cool yeah I'm imagining this like fear is proportional to one over distance something in that family of equations uh like a bus that's uh 100 feet away not scary 50 feet away not scary it's still on the flat part of the fear curve and then it gets exponentially higher when it gets closer to you and takes up more of your field of vision so um and then there's also longdistance threats somebody with a projectile or something like that and so that entails a causal model where somebody who's doing a certain action might be able to make an object quite big in your face very soon but maybe it's not quite there yet but it could happen rapidly so that's kind of an interesting take yeah so the point here is in along the line of what they were saying trying to understand what is or what could be some aspect of the causal role of Consciousness in things like appraisal and motivation of behavior here the point that we make is that actually the spatial structuring that we put forth based on projective geometry implies naturally something about you know relative size that is consistent with psychophysics and so it's to say that find somehow this Con structuration of space by conscious act as a sort of lens a magnifying device if you think about it that way that in virtue of its own geometry will have an impact that you know can then be used for other aspect of motivation and programming of behavior in the context of active inflence so maybe we'll get back to this Intermodal area again but that's the visual field and uh not all organisms or humans may have the visual field but it's clear that the spatial objects we visualize them in space and uh there's something similar though like if there's a lot of noise in the background but something is what you're taking up your field of audio awareness and so it's like in each of our senses there can be a sense like a sensory um expansion of focus greetings Lee just play on we're just continuing the discussion but go ahead there David yes so one of the essential thing we we emphasize is that that's it's actually not just

the visual field it's the C it is the cognition of space as in form in a multimodal manner you know through audio you may recognize object and all of a sudden you kind of have an image of sort the spatial representation of that object as if you could see it even though you don't see it blind people do that all the time they have actually perspectival representation and represent object in space based on other sensory channels there is an infer that is being made and the idea is that Consciousness operate at some level of abstractions once you have actually extracted information uh from different sense modalities and then they become spal object if you wish with variety of properties that you put in perspective in that field of Consciousness and based on which you will operate at the conscious level if that makes any sense yep interesting it's like it's so hard sometimes to disentangle from the eye eyes Vision perception debacle nightmare because we know that we're not experiencing what photons hit our retina we know it's a generative model and you're really taking it to the next level and it's like no you're not even just seeing a generative model of the visual field you're seeing this extremely inte you're quote seeing this integrated field of Consciousness and that um may depend on your visual abilities or your Acuity but it also something that's quite uh derived from Intermodal stimuli and uh yeah well that's cool stuff let's get to this projective geometry idea and we as non- geometers talked about it a little bit in previous discussions and just try to capture a few of these points that you said were differing critically between ukian and non-ukian geometries so I'm just curious and perhaps those who are not familiar with hearing about geometry and Consciousness in the same sent just what is it that brings geometry to the table and how do we end up even talking about geometry and Consciousness together excellent question so first of all a little disclaimer I'm not a mathematician even though since I understood early on that what I was trying to do required to think mathematically in Fairly Advanced way banged my head to try to progress but I'm privileged to work with real and very you know good uh mathematicians so that helps but nevertheless I can give you to the limit of my knowledge a some info about it number one I would say generally speaking uh you know if Experience One conspicuous aspect of experience of subjective experience is a spatial manifestation not all experience reduced to space but a big chunk of it is involved it so it's good from a scient ific standpoint because then it becomes something you can think about mathematically and a lot of models in let's put it that way in cognitive science has been based on information processing sometime very formal model like you know Advanced basion models or even active inference and so there is a lot of emphasis on the mathematics of information Theory so to speak and here the notion is that in order to bridge um information

processing if you wish with aspect of experience as manifested as a space geometry comes obviously as a scientific object as a mathematical object as something you know necessary so the the the challenge is then to understand how to articulate information Theory with geometry in order to have a fuller account of you know the process so then now on the U specifics of projective geometry it start from you know trying to find some let's call it that way phenomenological invariant that most people would agree upon to a certain extent considering experience introspectively it appears that it's obvious in Vision but I claim it's General in both perception and Imagination that the way we quote unquote represent the world let's not you know enter into the metaphysics of representation let's use it in a pragmatical way here this notion of representation uh is in perspective you know obviously a parallel line do not appear to us in Vision as parallel but they converge at a certain vanishing point toward Infinity um there is an horizon and it's three-dimensional it's not like just the projection of a 3D World on an image we experience it with a sense of depth and I argue that the same when you imagine yourself somewhere considering something aiming at something you know the notion of aiming at is very important in the phenomenological tradition in philosophy it's not just about visual Vision it's about spatial aiming and in all the circumstances the world is mapped is there a world that seems to correspond to surrounding a world that corresponds to thing we remember or even a totally fantasy world of things that may never exist every single time we build experience in Consciousness the point is that there is a perspective that is being taken with these properties of having an elusive point of view somewhat in perception we tend to feel it in we feel that we are in the middle of that space so that's the origin is somewhere in your head perhaps but that's not necessarily the case uh and that's not something that we can uh identify as an the way we could identify the location of an object in space it's illusive something that the phenomenologists have talked about a lot and it's in perspective so if you take those two things An Origin of the projection or of the point of view that is Elusive and a three-dimensional representation of space that is in perspective then the only geometry and the the geometry that account for that is projective geometry and here since we say the space is in three dimmension uh there is this notion of you know left right up down and and forward and backward uh it's a three-dimensional projective space and then that means that this space is ruled by the So-Cal projective transformation that are specific type of transformation that comes from a group of transformation that transform perspective into that space uh so for instance you can transform a point of view uh uh from the point of view I have here looking at that screen to an imaginary point of view I would have in a different position myself or

somebody else and all of that is naturally governed by that geometry so then that geometry appears naturally as a or if not D candidate to contr this aspect of phenomenological experience and therefore then you can start thinking about okay how that can play a role in the process of active influence nice thank you for the clear explanation um really in in that uh little bit the clearest part moving forward is this elusive origin and spatial perspective that we have if someone can imagine an experience that doesn't fulfill those two conditions then we're already breaking our assumptions but if you can that's great it's like write a paper about it but given that there's consensus that there's an elusive origin and a spatial perspective we can then say that we're in a family of uh geometries that are related to this projective transformation and so going from there there's other Downstream consequences so really difficult topic sometimes to talk about of course transdisciplinary ideas but I think it's really clear how the structure of your argument is set up Stephen and just just a couple of questions is is the is the um the distance as things go away is that and you feel less intensity in terms of the fear or something is that linear or is that kind of exponential or is it veryy depending on the space and and I suppose the other question is is the is is this transformation between one mapping to the kind of projective geometry it's so it's it's not like you store it in that form there's some sort of action to make it into this projection yes very good question I'm going to try to address them to the extent I can so number number one these experiments that shows the spatial relationship between subjective fear and distance they found is a power law with an exponent minus one so basically it's like the fear decreases as a function of the inverse one over the distance so it's not linear it's not exactly an exponential it's not linear and uh as it turns out that's the very basic rule of perspective when you want to build a space in perspective you divide by your coordinates XYZ coordinates uh by a parameter that corresponds to the distance along with you are basically projecting the direction of aiming that can be for instance in front of you so basically you find that in that geometry there is a similar operations if you wish as those that are implied by the the one of distance models that is empirically derived so that's nonlinear but it's a very particular type of nonlinearity that is not a super complex nonlinearity that can be handled computationally which I guess is is important and now for your the second part of your question let me can you can you remind me of it so that I well it's just about that transformation between however the brain stores it and this projective phenomenon that we experience yes so the idea here somewhat what we have to presuppose if we want to embrace that type of model is that there is an unconscious World model which is complex which is constantly updated that somehow store information more or less in memory uh in Clan manner

uh uh and uh uh and that we access that information by taking perspective on it through by transforming this sort of quote unquote a clan representation into a perspectival objective representation that involve multiple steps and that's something for instance we develop the most basic one I would say uh in our mon vision paper in which we show uh and we how the calibration of a projective frame me 3D so we set the parameter that defines the projection under free energy minimization that's the idea is free energy minimization acting on Geometry uh UNS Illusions such as the Moon Illusion explain illusion such as aim illusion and many other illusion we would not develop in this paper and that is somewhat the idea is that in you know sometime people told talk about access Consciousness you know let's not enter into the complexity of that debate but let's say that the hypothesis here is that in the way we access information that's in Consciousness there is somewhat a change of coordinates quot quote from a edian representation to a projective one and that operation is essential in what makes information conscious versus not you know model if makes any sense thanks for the answer let's go to Lee oh hi yeah so I was just wondering um to to sort of build on Steph's question um if if a fear response is uh exponentially related to distance does that hold or is there any research that shows whether that holds for a counterfactual version so I'm thinking here about phobia or trauma so if you have a particular experience you know often in p SD those things are replayed um so yeah so did you have any thoughts on that or any pointers it's really a great question uh okay the literature on on this on the the basic experiment is not huge surprisingly and this and your and your question is super interesting indeed because we would like to to demonstrate that even if you do it in an imaginary way you cover this type of effect consistent with the model I'm not aware of such study but I'm you know I'm doing so so much multidisciplinary work that the literature I have to cover is vast and sometime I may miss important studies in specific subfield but definitely that's you know that gives us the road map for a great experiment yeah in in the in some of the literature I'm thinking of um Benjamin Bergen on simulation he talks about the difference between um canonical perspective and I can't remember the other one but it's essentially firsters perspective um and and the ability to shift between these so I just wondered whether it might be related but I'd be certainly interested if there's if there's any research that you're thinking about doing yeah um David unmute and then go ahead and then Sasha so you want me to speak now oh if you have anything to add there no no noit sorry I misunderstood I lost the screen oh yeah no great Point um and thanks for that question Lee um Sasha um yeah very interesting stuff um makes me think about um uh this Consciousness and understanding of um how to interact with your

space as like narrative building and and how that develops over the lifetime that um like as a child that the things that you're afraid of are uh quite specific and nearby um and as you know as we grow up um we're afraid of much kind of larger but more distant things but uh there's something you know uh some transformation we have to figure out as far as um we still have to be motivated and um afraid of the things that are uh either very afraid or very or sorry very scary or very unlikely and those are still enough to motivate um people to avoid them and so I think that's a really interesting uh yeah something in there about uh changing the scope of your uh non-ukian projection let's say um as you uh grow up nice Sasha and it's kind of like when you go back to see something scary in childhood you go oh wait it was only a 6in stair that we were jumping off of well you're bigger but also you have a different worldview and then just like in the computational Psychiatry realm we might talk about difference in a statistical parameter leading to decreased motivation or decision-making it could be a distortion of this projection that results in things being uh perceived as dangerously close claustrophobia people are too close to me or people are too far away from me there's so much there with a social touch as well um let's go just on to the sort of new parts of 9.2 and talk about some of these other slides and questions we have so last week we talked about the functional features the cybernetic features as well as some of these other theories threads that were drawn in like the availability kind of like a workspace the simulation Theory motivation of action uh we talked about that last week as well as the questions about the phenomenological invariance with the emphasis on 01 being that relational phenomenal intentionality so just pause the video and look at those if you want more detail or of course read the paper but let's go to some feedback from participants so one thing that a participant who couldn't be here today asked was one thing I'm curious about is whether they're working on more complex forms of the model with higher temporal depth that kind of speaks to these adult fears considerations we talked about and then also this person added another thing I'm curious about more philosophically is what do they mean by now they have an extended phenomenological background for the model which uh suggests that the now would be extended like an extended now however in naturalist models which I assumed uh active inference or potentially artificial intelligence framework based models are a subset of time is often modeled and understood as point-like or as a continuous variable with points I wonder what the implications are or what there might be if they mixed up these two interpretations of what the now is the physical and the psychological now so David what do you think about these questions from the participants oh they are great uh the I'm going to try to address some of it of course and again it's there are still a lot of

things to discover and understand and one of the thing we like about starting with a mathematical model is that then it act as a guide now it's the theory that tells us you know where are the problem and where are the solution if you it's a general analogy sorry I'm going to be I'm going to I'm going a bit of track but it's I think useful and without cons comparing ourselves with Einstein don't worry we don't have that you know ambition uh if you think about uh general relativity once the theory was the mathematical theory was derived basically people looked into it and started thinking H there is a problem with this Theory it tells us that there should be black holes right and in fact basically the theory predicted things that you know nobody could observe of course before and very difficult to observe and that was a consequence of the mathematics and so that's the idea that we follow from a scientific standpoint is let's give us mathematics and see where it lead us now um one thing I'm curious about the the first question yes we are working on more complex form alluded to that at the beginning uh number one this notion of transformation of as an inverse law of distance related to this literature is not to say that we praise information only that way the it was to show at least to that type of relation hold in appraisal and to some extent or model makes sense of it now about the more complex form remember the space here the geometry is one thing but it's embedded into an active inference framework with a lot of complex computation involving prior beliefs and preference internal model of the body of others of objects including internal model that's what we are doing now of you know beliefs about the beliefs of others and beliefs about the beliefs that other belief are sorry beliefs about the belief that other believe other have so basically level six or second order theory of mind and by actually imbuing artificial agent with this type of statistical processing uh and combined with the geometry then we can start modeling how you know inspired from simulation Theory we can put ourself in the shoes of other we can try to imagine what it would feel like to look at things to observe things to might aim at things as they seems to do including understanding that they may have different beliefs and preference as us you know the same object may look cool to us and so when it get closer it's better but for them maybe that's something they hate and when it get closer they know they want to run away so then we have a way of extending the model very naturally with the same match with the same general principle to encompass theory of mind and social affective perspective taking if you wish and one interesting things that we find is that when you enable artificial agent with this type of process then they and you enable them also to be influenced by each other because you may be able to understand and predict other Behavior but not care about it but if you do then the agents the little robots or the virtual humans start Dem showing

joint attention Behavior they start interacting looking at the same thing looking at each other to kind of confirm that they understand what the other likes and they constantly put themselves in the shoes of the other and reciprocally and through this process of you know projective perspective taking uh informed by all these kind of ban like models under free energy minimization they start showing very complex Behavior including you know we have reproduced in a nonverbal way the famous San An task we know about the of mine basically that's two puppet one of them they are together there is some object of interest in a basket one of them leave the other take the object of interest from the basket and put it in the box and typically you ask children you know when the other come back you know where do you think you know she will go and if you understand false belief you will say well because she believed that the box is still in the the the object is still in the basket she will go to the basket and so we have model the kind of simulation framework in a nonverbal way for that and we show agent actually exploit their understanding of false belief in others to approach disingenuously uh object they would like by you know being freed of being buzzed by the other because now the other has false belief the same way we can kind of simulate some aspect of social anxiety because anytime you give you have an agent that believe that the other agent doesn't like it you know uh then you know playing the algorithm just running the algorithm anytime the agent imagine itself approaching the agent the other agent that the agent suppose do not like him then it put itself because he practice he does social perspective taking in the shoes of this o agent and when you imagine what it is like for this o agent to see himself or itself get close because of the same rule we have discussed he understand that would raise free energy in the other agent and since he takes into account the oent opinion that it attributes to the others therefore that raise its own free energy as a result avoid it tend to avoid approaching the agent that it believe doesn't like it so you see you start having very sophisticated behavior all through this basic uh processing so that's for the first question sorry it was long I hope it was not too confusing anytime you have to deal with theory of Mind in particular second order theory of mind and talk about belief of that another may have about the belief of another etc etc and so on it's it's become very challenging for the mind so now regarding um the other question about the now in the spacious present so in a way it's a hard question to answer and we are still working on that in a way U there is the physical time let's call it that way which is point like probably that is involved in the underlying processing know whatever it is based on the brain or based on some computational algorithm so there is a there is a a timer and then there is the time as it is integrated perceived and structured in Consciousness and the idea here

is that because we are doing active inference constantly uh anticipating the future to some extent even a very close future it can be at different scales so then we are trying to recover from that but we are not yet there the notion of this kind of extended present so that basically at a given time instant uh that's not the right at a given moment if you wish you not only have this instantaneous representation of space in perspective but you have a process in which uh the anticipation uh Drive the next instant in a manner that is integrated ated as a whole so sorry it's very abstract but I'm saying we're also trying to understand how we can put time pressure within this free energy context and we have solutions for that and also how the agents can dive into their autobiographical memory under the same process um we are also very interested it's now it's moving away from that but it connect to questions that we had before to the I would say pathological transformations in the sense that here we are talking about very adaptive behaviors very tuned to sensory motor contingencies and constraint and uh so that has this embodiment the fact that you know we tend to have this first person point of view refer to the representation that we have of our body in space that's constraint that embodiment constraint somewhat reduce the number of possible projective transformation ations and you know because only some of them will give you that type of point of views but if you relax those constraints then you can get very wild things that I find are very interesting to connect with the literature on psychotrop on mystical experience and arts and to give you an example we not published on that yet but one of the things that happens to the agents is that you know if they they have the possibility of basically being Freer in the application of the pro projective trans information they can use for instance in the context in which they are in a desperate world in the sense that everything they can think about or do as an embodied agent will yield a very high level of free energy there is no good solution then one option is to release the constraint on the projection and when you do that in order to minimize free energy the agents will tend to basically put their point of view quote and quote at Infinity what we called the god point of view which is a bit bad choice of words and so basically that means that one way of minimizing free energy for those agents is through releasing those con normal constraints is to escape towards Infinity so that basically the world become very very far and then the structure of the representation of the world you have start having very odd symmetries that may remind us of mandas or things like that so I'm shutting up now wow lot of stuff there let's go Adam and then Lee and then Stephen um hi can you hear me yep sounds good um so one thing I've been struggling with in terms of um time and Consciousness is uh whether it's better to think of it as like a series of kind of discreet snapshots or more of a continuous flow I'm personally

leaning more towards um discreet snapshots and so with respect to um uh projective geometric modeling I'm wondering whether uh the refresh rate um might be like something like Alpha frequencies like 10 times a second to uh primarily like posterior cortices getting into the mix of establishing that uh 40 perspective um but then more like you mentioned like access Consciousness earlier where that might start and like the whole active inferential unfolding that that might be more at like Theta frequencies like three to four times a second and then like you might get something like uh northoff style like a spatio temporal alignment based on like your interceptive States maybe more at like Delta frequencies and so you might have this like nested hierarchy of time scales with a core screening over events at various uh resolutions but at at the bottom of it though um it seems like where like the core of it is this establishing of a field um with projective geometric modeling at the center and I guess is this are these estimates I'm wondering discreet do you think or is it continuous that's really a very good question and very sound and interesting hypothesis you know with the empirical testing course it's it's a whole program of research I've been for long an electrophysiologist i work for long on this stuff I I don't anymore but you know I'm very open to pursuing collaborations on that um I first of all I should say I don't have an answer to satisfying answer to that question um I think it's indeed multiscale um I think that uh it has a discrete quote unquote sharp if not discrete component uh which may be perfectly embedded in a more at least in the way it manifest itself Continuous Flow but that is I don't think in of itself doesn't pose a challenge a theoretical challenge it's just complex but we can think about how it could be done and we can also think about how it could be calibrated based on the type of biological rym you are talking about what is more of a theoretical challenge is this notion of spacious presence in which somewhere that's the words that the phenomenologist like us would use we pretend pretention this notion in the now there is somewhat Us in the center of that now and there is at the same time a protention towards the future and somehow there is simultaneously as a manifold an experience of the you know here and now and an experience of the just after and at the same time they still this Resonance of this retention to use us world word sorry uh retention of of a immediate past so you can think okay there is an immediate past there is a actual present and there is a close future and then those things are somewhat bounded together but the difficulty is to try to think how actually you can find a mathematical expression of representation of that in which they're actually part of the same whole and not just you know something you could argue well but then that means that you know there are ultimately separate process so we still trying to work on that see if there is any way of thinking about that but we are not yet there

cool um we'll do Steph Lee Sasha yeah just building on on the idea that the projective Consciousness can be released when things are not certain enough to be able to know how to make sense of in the world it sort of ties in with this um way that indigenous approaches are shape-shifting and they're trying to attune to an environment which they aren't able to construct in the way that we tend to construct and place ourselves in the center of so I think there's an interesting thing as well as there being the risk just be the the increasing level of ambiguity and uncertainty in the world world so that's interesting that you take that I'm just wondering if there's as well as the projective thing going from the self and maybe to Infinity do you think there's like there might be another process another type of geometry coming in the other direction like it could be and and and and this it sort of shifts towards the self centered and there's an allocentric mechanism as well so that's tou a question and attached to another work that we are pursuing with mathematician Danielle Bena and KET Wily for who just told me unfortunately he could not make it today sorry uh about what sometime people have called prereflective self-consciousness it's a bit along the line of this bizarre sense of the spacious present but now it's about self-consciousness in the sense that that's what people have called sometime the self-consciousness hypothesis not everybody agrees upon that that we are always Consciousness is always conscious of itself that is it's not like a higher order Theory where there is a Consciousness for instance perceptual consciousness of something and then you reflect upon it and you know become aware of the fact that you are perceiving that the idea is that it's actually the manifold itself is such that you know there is this pre-reflective SL reflexive uh relationship that is fundamental into it and so philosopher like SRE I believe and others I've talk about the how to say that in English the um the seers seeing or seen the the seen seers if you wish the the one we seeing is also seen at the same time as if there was some sort of duality in which whenever I have a first person perspective and I'm looking at the sky it feels like or there is something in it that feels like that the sky is looking back at me well as it turns out it's a natural property of the most you know bizarre strange and interesting aspect of projective geometry which is not just you know it's a very complex and Rich geometry uh and we're trying to work that out thanks to this mathematician around this concept of Duality reciprocity uh in which there is at once potentially we still working out the ma uh uh this the fact that you take a perspective in implies that you know there are reciprocal perspective and that you know you look at something and these things these things look back at you uh in your I know it's odd it's bizarre but you know it's my my attempt to try to touch upon what you are talking about okay Lee then Sasha Adam thanks for all these cool questions uh yeah so um I'm

anticipating that I might be about to ask a stupid question but I'm I'm going to go ahead anyway so it was um it was stimulated by the fact that you were saying that you were kind of interested in you know like mystical uh experiences in relation to projective geometry and and I was thinking about the um uh the meditative experience of of of nonduality um you know which is um I was wondering whether you have come across this and whether you see it as in any way related to um uh projective geometry perhaps yeah so I used to be I mean I was at some point I don't know if you know that person a student of Francisco Varela unfortunately passed away during my PhD that's why I moved to the US to work with Antonio Damasio to finish my PhD and uh of course with Vela who was very interested in let's say Meditation in a broad sense of the term um both for a scientific standpoint from a scientific standpoint as a possible method to study Consciousness but also more at a personal and existential level you know I got more acquainted to that I practiced it even studied it a bit uh and I've been for instance involved in a very nice Workshop in Israel and perspective on Consciousness where this type of tradition multiple tradition we are meeting together and we had very interesting discussions and thought and about about that very nice group so um can you try just for me and for the audience perhaps to specify the phenomenology the the aspect of the experience you are describing the meditator the specific meditation experience that you are describing this kind of unified experience sure I I can try um so I think in in some sense there was um uh prior to a certain point there was an assumption that there's a separation between the Observer and the and the observed and and that starts to break down such that you kind of it becomes it becomes one one space for want of a better yeah so that I think that of course I'm not saying that's the answer I'm saying that perhaps that contributes to that type of state that connects to the what I was talking about of that intrinsic to this the reciprocal relations that are embedded into this geometry I wish the mathematician I work with was here uh there is something like uh it's nontrivial that's important a reciprocity between looking at or aiming at and being looked at or aim at simultaneously it's like it's not like necessarily you lose track of where you know of of the perspectives that you have but there is this more the Richer experience that is implied by this type of geometry so in that sense in that sense at some level you know you you can perfectly conceive within this geometry how you know perspective can be taken from anywhere and are all connected through projective transformation but even deeper uh perhaps through Notions of dualities that I unfortunately could not expand upon because I would you know totally fail that's why I need to work with mathematicians uh that gives you this sort of you know unified experience of a you know sort of some sort of bizarre panopticon where you

you know you could be anywhere you could be looking from anywhere and from your particular perspective and being looked at that could be you that could be something else and that's you know a manifold of point of view that at the same time is grounded into a unique point of view that exists in full Dimension so I know it's sounds Des but that's just the math that's where they where they are thank you oh we have a new visor but Sasha go ahead then Adam um yeah uh I'm just thinking about this whole time perception question um whether and I guess it's it's two separate things it's is time discrete or continuous and the second question is time perception discrete or continuous and if it is possible for uh people to have different you know spatial uh understandings or perceptions then um it's also possible for people to have uh different perceptions of time passing and uh different rate of how everything is happening and is that um how do we integrate that or allow for that to happen that people perceive time uh perhaps very differently well that I don't think in and of itself um is deeply difficult conceptual challenge at some level at least uh in the sense that it's not different if you want to think about it as having systems that have internal model of beliefs and understand in the that you know somebody else may have different beliefs in my robot in the example I was giving it was many beliefs about whether or not an agent likes or dislikes something but uh uh uh you know you and all this let's put it that way let's be broader in the mechanics of the whole system there are a lot of parameters uh that concerns beliefs uh that concerns the the calibration of projective frames that can be different from people to people so I would guess that it's not a satisfactory answer I'm sorry but that could be the same for the the way the the flow of time is U uh structured is is construed uh within Consciousness is to me you know at some point you are living with complex systems that have you know statistical models with a lot of parameters so it's perfectly logical that if you have parameters pertaining to time Consciousness uh they could be different from one person to the other you know at that level of abstraction I don't think the the problem is a is a an essential challenge in the sense that oh here there is a big Paradox that I cannot resolve in a consistent manner with mathematical thinking but of course there are much deeper questions and this whole what we were talking about for this whole spacious present this kind of extended experience of experience of the present of the now is more difficult to grasp to me yet let's go last comments here from Adam and then we'll welcome Ken hello can you hear me yep um so and talking about um this sort of reciprocity or mutuality of like when you look and then looking back at you it it started to make me wonder about um the extent to which uh projective uh geometric modeling might be uh inter subjectively bootstrapped and that like it both enables um theory of Mind capacities in terms

of giving you this flexible transformation between perspectives which would be like at the heart of of like mentalizing a theory of mind but also to engage that I'm wondering if that helps to induce the capacity which then feeds back and makes you better perspective which then and and the kind of bootstrapping process and now I'm wondering in terms of like Evolution and development like when is this starting to arise like you could in theory think of this happening like nonin subjectively in a way where it's like I don't know like a predator sort of like hunting maybe if I go here and doing sort of like where do I need to be to get like if the organism has like a model of itself of some degree of temporal depth and it's trying to do something a little bit more labyr and trying to like Forge for information well that's go okay I go here I'll have a different perspective and this is like influencing the policy selection but um in humans it's probably really heavily inter subjectively bootstrapped and know so some thoughts I'm not going to make a long comments to not take too much space but I find everything you said I agree with you I think actually you right probably so uh there are very interesting stuff in ethology and you know animal behavior and even in recent relatively recent literature that you know suggest the role of projective representations into the way animal try to lure others into you know either being a ma or you know being a a pre cool welcome Ken you can say hi yeah sorry I'm so late all goodful morning do you have any uh thing to add or just feel free to raise your hand any time okay we'll do cool thank well there's a few more slides this was really a great set of questions by the participant so that's really fun topics uh here's some quotes that I put up there and the part I wanted to kind of start off with is the second half of of the uh second quote on here which is just uh a system running the PCM would more closely approximate the actual complexity organization and processing power of human or animal brains and this could very well on the author's view literally be an artificial Consciousness so maybe getting into sort of questions about what does it actually entail to have the awareness what if you make a system that has action and perception and it does a projective geometry transformation and it you know it's doing it because you're the one who made the robot so I was just curious to hear about from the robotics side or from the mathematical side or from the neurophenomenological what else is happening in humans is it evolutionary is it linguistic Etc so I I would like Ken to to speak because he he has a lot to say uh but since you ask things about Robert in that question is very relevant of course I will answer as a scientist I'm hiding be behind the method the scientific method not to avoid the question but to tell you first that I don't have an answer to that fundamental question what I'm trying to do and that's I think the only thing science can do at least as we know it and you can

take the model of physics as a prominent exemplar of that is to try to understand not necessarily why but how we do think we do things and that applies to Consciousness to the phenomenology of Consciousness so what we are modeling here is that phenomenology and how it is involved in other process and so at the end the only thing I could say about my robot in the way they are implemented at this point you know there is a lot of work ahead is that at least their behavior will be governed by principles that are Central to the way Consciousness process information let's put it that way so it doesn't imply at all that those robots are actually conscious in the sense we we would like to um but at least uh uh they can demonstrate how our Consciousness and the principles that seems to shape our Consciousness play a role in overall the economy of our Behavior etc etc it doesn't give us a an answer to why you know this is conscious in the sense maybe actually it's with a certain level of complexity sufficient and maybe Ken should speak now uh and rebound on that well I was going to say this um passage if uh memory serves we we we of course this is a a general sort of worry that anybody who uh works on Consciousness thinks about it has especially if you are doing anything with u simulations as David was so we started to think well of course we don't think that um the simulations that David has built are actually contained Consciousness you actually contain agents so then you wonder all right you know so it's not it's not merely um processing information in the the way in a way that simulates what's specified in the model I guess you could say um it could be other things as well like uh so in a sense we we think okay this is a necessary condition for Consciousness that you process information and the way the PCM outlines um sufficiency require a certain kind of uh scale and complexity that that we we're not we're not certain about yet so we shouldn't just um assume that any any kind of that that uh follows this this model is just ifof facto a conscious being that's kind of claiming too much what what else is involved well we we don't we don't know but um it it seems like uh this is we we we maintain this is necessary for Consciousness so um conscious being will will have this structure well will process information in this way um and for all we know it it could be sort of um things that we don't we're not going to be able to reduce to uh principles so easily that we'd have to discover empirically that's the that's the reason we say here the an aposteriori identity Theory that's that's the idea that you can't really say what Consciousness is AR priori you have to find out what it is uh empirically and at some point you might find yourself in a situation where you could say Okay um these suppose we build a lot of artificial systems that do this these exhibit all the behaviors and so on that make it seem like you know a conscious being um they you know suppose they operate on a certain scale and a certain complexity at

some point we might be able to say yeah we think right here's The Sweet Spot why exactly we don't know maybe we will discover that maybe not but um on the one hand uh it seems like you don't want to be um shy of saying that the you know from what we can tell these are necessary conditions and on the other you don't want to be to um you don't want to you don't want to have a view of the sufficient conditions for Consciousness that make it quite too easy you know so awesome yeah thanks for that response Stephen just a really quick comment and then we'll move on just just quickly I was just wondering when you think about projective Consciousness is it like this is really summing up a particular type of Consciousness which that we we are taking these projective perspectival kind of um ways of engaging the world and there's there's a potential I wonder if there's a potential for that to seem like it's more like how Consciousness is because we culturally have taken that on to be how we think of the world but this might be like one way of taking a view from a more sort of higher perspective so should shall I take this one can please do and speak more what's that I think you should speak because I have spoken already a lot and you so I mean our our hypothesis is that this is not just of um you know uh how we've been enculturated or uh let see if I I I got your question right I mean r view is that this is this is a at least a necessary feature of all Consciousness that's that's the claim so now you could imagine and you could you could actually wed this kind of view to other theories of Consciousness you could wed it to you know pans sism for that matter you could just say oh no actually Consciousness is something else it's whatever some fundamental property of nature um but this then if you had that sort of view this kind of work would still be of interest because it might um sketch out how cons ious beings like us or probably you know animals um how they experience the world and how they organ how they process information all right so but and of course when we make it so it's compatible with that comptible with um really of Consciousness that's cast at a kind of metaphysical level I guess um our rvw would be something like well you know suppose this suppose this type of model ows us to explain everything that we we think is pertinent to Consciousness so it allows us to account for whatever behavioral and cognitive functions um Consciousness carries out and for the phenomenology of Consciousness um if it if it allowed you to to count account for those things um then and you suppose you could identify the type of structure this is and the the sort of scale and complexity it needs to be organized on then to our way of thinking at least as it comes to the metaphysical issue we're we have a tendency to think okay that's just what Consciousness is we we discovered this empirically we discovered it aposteriori there's no need to postulate any further metaphysics because any any metaphysical account of

Consciousness will actually has the same structure it's it's something you didn't figure out a priori really you just thought okay well here's a here's an account that would make sense of the fact that you know say take pan psychism why is anything Consciousness well you conscious well you know if everything is conscious then it's no mystery that we are right so so that's uh so that's that's an argument you could make well you know as you as you probably know there are all these different um positions in the philosophy of mind about the nature of Consciousness and the and the distribution of Consciousness that in my view they're all kind of epistemically On a par none of them are really like you can't show any of them a priori you can't demonstrate so um it seems like it's wide open to you on the theoretical level to be this kind of identity theorist just when you reach you know when have this structure have this level that you're a conscious being that's what Consciousness is we figure it out empirically and we don't need to sort of postulate anything else that said it's totally compatible with with other views so you could hold that not this is much more derived and primitive like sensing or something is is a basic form of con or something like that um we we do think though that e so here's one thing that's it's not exactly AR priori but let's say it's a a strong intuition that a lot of people have if if you're conscious of anything that's going to involve the appearance of something to so it has this kind of appearance of to structure and that's a kind of spatial spatial I mean quasi spatial structure a protog geometrical structure um and and that takes you already kind of in the direction of of thinking of Consciousness as like a space and like a projective space in fact right um that's not you know this you can't demonstrate that a priori but it is a strong intuition that people have so I'd like to say I think I say in this paper don't I um you know follow your phenomenological Muse build your build your models and see where they take you because we can't really determine those things a priori or just by banging on the table with our our favorite in thanks um I I see Adam and Lee have their hands raised but just really quick would uh especially uh David and uh Ken would would either of you be okay with staying for like a few minutes past the hour so we can just continue talking a few minutes okay I have I have a call in about 40 minutes but so I'm fine oh beautiful so we'll just um anyone can depart whenever they need to and uh otherwi we we won't have to come to a speedy Landing there but we'll go Adam and then Lee oh hello um I guess I'd like to hear a little bit more about uh the intuition for or um why it is that existing simulations might not be conscious and in terms of um what it would actually take to uh get a sufficiency um this is something I've been struggling with also um the uh for me like I've been like mostly coming at it from like the neural Direction thinking of like okay what in terms of neural systems might

underly these different capacities and then like what uh how would that be implemented um I become a little bit more confused when I start to think of alternative implementation so like something like um uh projective geometric modeling I'm wondering if that's primarily like a posterior medial C posterior medial cortices and that's like giving you this sort of flexible spatial modeling but then this is like coupled to like um body maps of like lateral parietal lobes and that this is part of like one um basically synchronous complex which is mutually constraining each other as you're filling in this sort of joint estimate of the world and your lived body in relation to it so basically exactly what you propose um and so for me it's like when I'm talking about a biological system there's a certain grain at which this calculation is being established through like a metastable synchronous complex or might call it like a non-equilibrium steady state distribution or I call it a self-organizing harmonic mode but if we're talking about something like a simulation on a Turing machine um that might be doing the geometric modeling by very different means um is that an issue is that not an issue for you uh just curious to know more thoughts on uh boundary problems thanks maybe I can give my thought about that and then Canen number one about the functional neuroanatomy and the neurodynamical mechanism that you know may implement the type of process we are talking about I think agree with you in the paper we published in 2017 in the Journal of theoretical biology we have at the end a a little figure that kind of summarize or gross functional neuron anatomical hypothesis and indeed it seems that the posterior you know in particular the and to some extent the parietal cortex but the parietal cortex may be more is a bridge towards interaction with other system but the parietal L and then we could enter into the detail uh uh is certain is probably where this sort of project as we call it in the paper projective geometrical engine operate and I think the way you attempt to make sense of the difference between you know something like the model being processed the way you describe it into a brain versus into a Turing machine basically a computer is is perfectly fair and right and maybe indeed when you do it in the Turing like way like when I do my simulations with my virtual humans and robots you know it doesn't you know until the type of experiential aspect that we are trying to figure out of which we are trying to figure out the cause basically the sufficient condition and perhaps in is it's a very different context of processing the way you describe it and that is actually key perhaps not you know it would need more elaboration of course there's still explanatory gaps here but uh to make that process conscious so for me it's not a problem it's different type of uh of implementation we can use the Turing life to do simulation from which we can learn a lot but probably indeed there's something about you know the embodied

processing with this type of what about the brain is doing and you know I'm a neuroscientist initially and I can I we all know that actually we we know much less about the brain than you know we pretend to uh and there are probably great and important deep fundamental principles still to understand and until we have understood the brain in a much deeper way uh um well we we won't be able to con on that question Ken yeah I I can just D that you know um I mean suppos it it's it sounds a little odd you know from the kind of computer science perspective to think that say doing performing all these computations in a massively parallel way so that um you know I mean you can imagine you know a tinker uh computer doing the computations involved and taking like thousands of years or something something and you think okay that's it's not it's not merely running this program right but if you say okay it has to do with um massively parallel integrated computation of these uh functions um that of course sounds a bit like um like strange like why why would simply being able to do it really fast or do it in this parallel way make it make a difference um but you know if if you're serious exterior identity then you think well okay we'll just find out right it it does it does seem really strange though to think um that a touring machine made out of beer cans and string as John Sur carrying out these would be Consciousness so um so that's why I I think it's it's a good idea to leave leave a little bit of SP here right so that's which is what we to do maybe let me just add a little historical note it's not about touring but it's about vanon nyman you know in his seminal paper in when he was at the Bell Laboratory I believe and you deriv he explained the foundation of his architecture that is now basically a an implementation of a touring machine into the computers that we we use currently um actually and then you know it becom like oh you know vanman Machine versus the wet brain but in fact Van Norman in this paper start from models of artificial neurons the basic mular canit neurons to try to figure out how basic computation such as you know arithmetic operation could be abstracted and then he proposed a Serial model of computing emphasizing that at the time because they had to use those vacuum tubes to make those the equivalent of semiconductor you know it would be too too big to build a parallel machine it would just not be physically practical you would need like a giant building just you know to put enough tubes in there and so the solution of doing it in a binary Manner and in a kind of sequential manner it was just a practical solution okay but his thinking was broader so you know maybe there are also deeper aspect of computations uh you know we don't understand yet and that could help us also to make sense of the experiential Dimension that's an hypothesis and if I could add one thing that it is also true that simply massive complexity hides I mean it it has a sort of I don't don't take this too seriously but you

all know the old sort of Hegel's idea that quantity eventually yields to Quality so you have quantitative changes and you have enough of them it's as if it transitions to Quality well there there is one way of thinking about that in a kind of not not taking the metaphysics of dialectics seriously but looking at it uh epistemically where very very complicated things can end up being in fact this is kind of the way we are uh oh somebody brought this up okay okay well all right so I'm speaking to nobody's gonna be freaked out if I start bringing up go know um anyway that the idea is yeah that particular notion of of uh going from quantity to Quality so just think of massive complexity you take a mass complex system as a as a as a whole it can behave in ways that you you have to understand sort of qualitatively um you can't I mean you know the the even things like CH chess play computer programs to take the classic example approach your chess board playing computer if it's a good program from the intentional stance Dennis term you you regard it as you know trying to attack your and so on you don't like think of the low-level complications going on in it to to implement that so that that's all just to say massive complexity with the parallel massively parallel it it's not totally crazy to think that that um could yield something that has this kind of form and this kind of macro structure if you like that we experience qualitatively that Consciousness has it's not completely crazy to say okay yes serial computation couldn't do it but massively parallel computation could here it's no crazier than a the of Consciousness yeah go I've been um thanks for that I've been um that makes a lot of sense to me and I guess in my own thinking I've been struggling with like to what extent I'm dealing with practical versus in principle uh uh issues with respect to like boundaries and like what it would take to realize Consciousness so like you say like the massive parallelism is practically necessary but is it in principle necessary and like and there does seem to be something strange like you know Church turning uh thesis like in theory you could do it with the beer cans and rope and let it just play out in a really long time scale and eventually you calculate the whole darn thing right but you're just like I don't know about that but but why so it would be like well practically there might be like a sense where like okay you need Consciousness to both like inform and be informed by action perception Cycles or something like that but that's still a practical not necessarily like an in principle but then like I I guess I've been wondering like okay so if it's we're going to in principle would be like is the computational objects like real or not um so like uh if like like like a simulation versus an emulation or things like this so like in like like in with like the recurrent Dynamics sort of like metastable synchronous complexes model I'm working on uh even then there's like an issue of like is this a real thing or not and like they're dealing with

like this issue with respect to non equilibrium uh steady states in the free energy principle which is like these um these things are defined as these like harmonic modes um but then their boundaries are like um they seem to be um model dependent like like what um or like an neural system like do you ever get like exact synchrony how much is enough to count um what's the uh what's the time scale at at at which the thing actually has its existence as a particular attracting State and relative to what and so this is where I start to uh yeah this difference between practical and in and in principle issues of implementation and what would actually and then what would let you say okay this thing is conscious I'm I'm still really confused about it anyway may I interject well ju do me just say may I comment on that yeah I'm gonna go if you com then you I don't seem to have a opposition so I started developing an hypothesis a while ago um that I published in you know in a journal called advances in Neuroscience that actually died it I don't think it still exist uh and it was kind of based on a reflection on my own work and the work of other in neuroscience and neuro psychology notably working with uh brain damage patient showing that you know a lot of what we believe as being critical for cognition in many aspect of you know cognition and effective processing in the brain like everybody believes now that the insular cortex is critical for craving and know many aspect of the sense of self well I worked with brain damage patient that basically had no insul on both side lack 40% of their brain and nevertheless you know except for some deep anog Amnesia that could be explained by their whatever uh uh they were normal so I came to actually think that the brain is much more complex than we think that there are the way we observe it with the method we have no Imaging electrophysiology is very low level it's not different than putting omers on the microprocessor and trying to infer what type of operating system running and so I started thinking so maybe there are you know layers of abstractions that we haven't understood in the way the the brain process information and one analogy that came to my mind I was very interested in computer science at the time is the process of virtualization in computer science that is basically a way I mean it's a simplistic way of explaining it but uh a way of creating abstraction layers so that actually at the user level of course we're talking about computer involving user but at the process level uh uh all the details of the hardware are hidden you know when to take a very trivial example that is not a good analogy when you use your desktop computer you don't see the the the machine language you are not using word you don't need to program in Assembly Language to use your Microsoft word uh interface so virtualization is actually a process through which um software or process can be uh abstracted hide the complexity of the hardware uh and also become much

more resilient in the sense that then you can kind of move things around across servers across you know different things so it's not an answer to the question but I I still have this intuition and believe that perhaps there is something analogous that is Bridging the Gap to some extent between you know what we know and observe of the brain and subjective experience and more you know higher order level process that is analogous to virtualization that basically the phenomenal experience of Consciousness is very useful for active influence Etc but relies on neural mechanism that precisely are designed to uh hide the Nitty Gritty small and complex details of the hardware and there and therefore makes it more you know usable more practical and of course at the same time it takes longer because you need to run many cycles to do that for instance if you want to S simulate a certain type of microprocessor over another architecture of microprocessor etc etc so I think there's something there about trying to understand this how the brain May create virtual space and through virtualization process that then may help bridge the gap between the different domain we are talking about even though it's not sufficient yet in my mind it doesn't fully address the problem just a couple notes to um tie together a little bit of the discussion so we've heard about a few classic philosophy of science stoies like necessary and sufficient uh practical and principled from philosophy and then also like real versus unreal seems like to some degree brains would like have to because they're so complex they would have to create these abstraction layers just to like work as Control Systems just to work as Integrated Systems they'd have to yeah like you'd have to like self core scin just to actually have purchase to do like heterarchical control and then once you're doing this um then you might get potential like flexible multiple realizability from that potentially or or or or non flexibility too depending on like how you did it but yeah you summarize Adam again very good summary of in better presentation of my home thinking well you've uh influenced me very heavily so uh I don't think that's a coincidence um yeah one just to bring a few of these threads together because it's all really interesting stuff so one is that in complexity we say more is different and that's related to this qualitative to quantitative uh more quantitative changes can lead to qualitative differences through phase transitions self-organization just qualitatively different Behavior strong or weak emergence from just quantitatively more and then that's always going to be a relational question how many grains of sand are in a heap that's the whole point is that's not pretending to be an objective question it's about object but not pretending to be a non-relational question even though it is about the world uh and these ideas about which levels of the system are available for introspection versus is related not just to attention schema Theory and attentional salience from the

Fe and active inference but also a lot of things in girdle leer Bach and so there is a a hoffstead citation in a lot of these ideas have a lot of analogy there which just shows how in some senses many of the same questions are returned to even 40 years later so we have to balance out the big Computing hype from the really quintessential philosophical questions like necessary sufficient practical versus principal real versus unreal perspectival and taking that seriously the relational perspective integrating topology and geometry so it's just really awesome to hear this great discussion and that the way the paper and actually the sequence of papers is written is very clear interfaces and highly compati focus with different metaphysics but just saying it's neither uh simply necessary nor sufficient to have this projected mapping it's just under condition X and Y for example The Vanishing um you know privilege point and the all these things we were talking about it just goes Downstream from there so it's an interesting approach and as we're hearing from this discussion a lot of that come together in thinking about these topics so thanks for like bringing all these perspectives the tables the school stuff um maybe Ken can comment on your reference that you had before on the glass of St works and get and bar because actually Ken has worked a lot on these questions has actually contributed to the I'm a strange look strange Loop book of the glass of Stater by editing a version of it and he knows very well maybe he can things to say cool this is one the this is the section that where it was one of the sections where hoffstead was cited but I'd love to hear about this Ken it' be great thanks yeah well it's just a little bit of U I guess history about this like um actually I met David through um he he and Antonio deasio were contributing a paper to a book I was editing um and Douglas hofstead also had a paper in that book and um you know the papers were supposed to be be like I don't know 30 pages maybe and hoped or turned in 100 pages so and it's kind of typical way so I I uh I said well look you know we have to edit this and he's like you know um okay but you know be very careful and so I went through this long process of editing down his contribution I cut it in half and and that eventually became um I am a strange Loop which is cited here um but he he was uh ultimately okay with with my editing job but that's yeah so and and in particular I I what I thought was really ended up being really nice about um I am a strange Loop uh the way in which Hofer is able um describe you know the ca basically the causal efficacy of um imple you know of implementing mathematical forms and so it's it's it's you you you know you T at least in philosophy think think of mathematics as a a totally like abstract platonic way it doesn't have any causal powers and so on and then of course there's the old much you know there is the Aristotelian tradition which says no no the forms are actually in the matter they're

implemented in the matter and but that's that position is not particularly um I mean you can find it but it's it's not kind of like it's not the dominant sort of thumbnail sketch of this issue in the minds of most philosophers but I found Hof's book to be really um really good on that like you know implementing certain computational structure certain ultimately mathematical forms has real causal power you know it does change uh the system that implements it and and and it's perfectly legitimate think of it as a causally real uh thing yep one example from hofer's writing that reminded me of today was when we were talking about whether it would simply be enough to have a data processing model that involved a projective geometrical component would that be necessary or sufficient for Consciousness and your very articulated answer was about how actually we know that these are some sort of uh features of what we want to be modeling but they're neither at the necessary nor sufficient tier they're things that we're seeing as like sort of being spun off from some underlying process that there's still a lot of mystery around and so that's like the phonographs um with the low resolution photographs in go to leure Bach it's like you can make a very limited thing that does one thing without error or without some of the fancy attributes but then to extrapolate from this less is different more is different case it's like well it's not a phonograph it's a hammer it's like okay it doesn't play records yep well it's that kind of a phonograph that's what it's like to be that kind of a phonograph that's what it's like to be a light switch okay okay well that's not really the complexity of system that we're talking about here which is the embedded in culture human and then that level of complexity the full featured phonograph opens itself because of its strange Loop nature to this whole new universe of recursion and of loops and of uh self-reflection projection and then the phase transition or more is different question is like how do we get from things that don't have that strange Loop feature to things that do but there's nothing that is sort of a strange Loop or is not a strange loop it's like either it Loops back with sufficient complexity to be strange in the world or it's just not strange enough to be as interesting as these open-ended infinite generative systems that we seem to be participating in Adam and I'm curious about how this interfaces with free energy because I I don't think there's been a hset or free energy crossover quite like this Adam hi um I'm thinking again about what uh David was mentioning with this sort of like reciprocity or mutuality of perspective of like seeing and then like looking at something and then this and being looked at and the ways in which um some of these things might be inter subjectively bootstrapped and I guess I'm wondering um whether that puts like a strange Loop like right at the heart um of this model and I'm also wondering whether one of the things that might like potentially get you is um

recursion and like and a a powerful kind of reflexivity which would then you can then start talking about things like conscious access and all sorts of metacognition kind of like growing out of that potentially um yeah some thoughts K you need to answer that question you know in reference to self reference and nonwounded set theory and whole project Etc you are the the one to to address that okay well so actually that volume that David contributed to and deasio well deasio but Heder contributed to it was it was a book called self-representational approaches to Consciousness um and it's was most of it was written by you know analytic philosophers and some some couple of phenomenology phenomenologists but the that was the theme basically how do you how you know people who think that like like I long have that Consciousness involves this kind of reflexivity kind of self-awareness that is not just it's not just introspection it's a more basic notion of reflexivity than being able to attend to yourself or reflect on yourself and that that book was was about that but uh in that book I kind of looked at ways of formalizing this using so the Hop the hopare we we I asked Heder to contribute to it because we we thought that's that's right in line with ger leer Bach and the notion of kind of using the notion of metamathematics and self-representation in self reference in um in the metamathematical context as sort of one model for the kind of selfawareness that's involved in Consciousness he of course that's the Str that's the kind of quintessential strange Loop for him with respect to Consciousness um but we I you know looked at models using hypers sets like sets that contain themselves or as members um and now recently though I mean we we kind of actually working on that is kind of at the origin of the the whole project the whole PCM um project because um I I moved back to Iowa City at in like 2007 I think that's right and um David was there um at uh still in the in the he had um graduated from the Neuroscience program there he was still working at the Neuroscience program at Iowa and we kind of formed a little research group just based on this and this eventually you know sort of we we could of discussed geometrical models a little bit but but David really had the Breakthrough um with regard to thinking of projective geometry as the the the Way Forward U somewhere around 2010 or something like that but we um we long wondered how how can you you know can you really connect these two threads this the the geometrical part um the projective geometry and um this notion of for lack of well to use Hof's term strange strange loopiness of Consciousness um and we we're actually still working on this so we we we we uh we think we're not certain but we think we thanks to Daniel because D Daniel benan um we think we we found a way to connect these two Notions finally and more clearly um D do you want to say a little bit about that about so how the how the how the PCM can sort of um make sense of

the this notion of strange Loop loopiness strange loopiness yes so that was really a driving force in in trying to select a a model to that the model would incorporate intrinsically properties like that so when I was trying to because I always had been convinced that you know there was something geometrical about intuition and so we needed to move from kind of abstract graph Theory and even with this kind of non well funded sort of set theory that allows you these Loops Etc into something more phenomenal more experiential more concrete and so I was looking into kind of possible geometries that had weird features uh Beyond this perspectival intuition from which we introduced that and this projective geometri geometry uh under some boundary condition but actually it's intrinsic to it as a natural Loop if you wish what I'm going to say is going to sound very awkward and it's a kind of qualitative description of things that needs to be expressed and formulated ously so sorry for to the mathematician that may listen if I'm not doing it the right way but basically when you look at a point at Infinity in front of you okay so the vanishing point just in front of you that's the same point that's the back of the point that is actually the antipodal point in the back antipodal points so two opposite points at Infinity in that Pro in projective space are topologically glued so that some sense actually if you were kind of a virtual little creature moving into the space of Consciousness uh when you reach the point at Infinity you find yourself on the other side of the space uh and uh if you think about it it's just as if uh actually the projective space you know at you know this when you kind of start messing with the or considering the plane at infinity and it's top ological properties basically was kind of a sort of mirror in the sense that basically when I look in front that's not the case in your visual experience obviously but that's in that is in the structure of the geometry I look quote unquote of myself looking in front from the back in a kind of reverse way that's a property that is actually intrinsic to projective spaces but that we don't normally see okay cool stuff here thanks for sharing all this history too so just two thoughts one is this virtualization idea and in gerle user Bach it's Prelude ant Fugue Aunt Hillary the ant colony and uh is it a virtual machine well there's yes and no in one sense lower level details are shielded to the more semantically operative top levels so that's virtualization for sure concurrent programming as well but is it a virtual machine in the sense that it's a VM that I can run on a different operating system no that's not even the ant colony OS that's not a strategy that transfers so the virtual machine metaphor captures one aspect perfectly which is that there's shielding of lower levels of the quote assembly code but also there's another feature of virtual machines that they've come to mean in the modern Computing environment which is like reproducible Computing and that's definitely not the

case that's because we have this embodied inactive ecological element so that's kind of a cool uh extension of a hofstadter idea and then this idea of infinity being on the other side it's sort of like you would if your light rays were being bent around the sphere or uh whatever shape we're on it would be seeing around so that's straight out like a laser so that's sort of an intentional laser reflected by the higher concentration of photo receptors The increased sensitivity to observations there um but that being the point of attention and even meditative practices often focusing on attentional points and uh visual Rays like that but it does if you think about the the ray of light going all the way around it would be the trajectory you would look at to see yourself from the back so very interesting there and that's related to our conception of the space of the earth whereas on a flat plane it would just go to infinity and then pop through in like a piercing way because the space doesn't Bend back on itself Adam sorry yeah so I'm thinking that you um might be providing an additional account of some of the uh phenomena that uh gratiano tries to describe with attention schema theory in terms of um certain intuitions selfhood and a kind of like um awareness is like a soul like ethereal non-material intuition and so for gratiano for him uh Consciousness is you know it's this low dimensional sketch necessarily reduced dimensionality for the sake of control um to have a model that's more handleable that throws away details and gives you purchase but that this like low dimensionality to it is like the thing that makes you that if this is your awareness this is um and your sense of Consciousness and and self coming from this that this is part of certain intuitions about like immaterial souls and things like this but if your perspective inherently involves this kind of like strange Loop this like difficult conceivability where it's like you are both the constant context for everything the reference point but you're the one thing that's like invisible and in this kind of paradoxical way I'm wondering if even I'm wonder if that potentially contributes to similar aspects of phenomenology where like certain asp of kind of like mystical intuitions about the self and Consciousness some thoughts cool question and we're taking it nice um and interesting areas David why don't you go ahead Ken Ken could say a lot of things oh sorry somebody else wanted to talk nope you or Ken go ahead okay I'm G I'm gonna just say something but Ken will be able to better elaborate there was a if a Mystic or say zist named Rudolph Steiner right K who actually uh believed that projective geometry uh because of this sort of reciprocity of perspective when I look at the sky the sky look at back at myself was a connection to God we don't endorse that view necessarily but there is an history of mystical and esoteric Traditions that invokes projective geometry in the past in that kind of context right Ken yeah so there's there's the Steiner uh tradition which um one of his students I

believe she was one of his students Olive Witcher she wrote a really intrusion to projective geometry um which uh I highly recommend looking at it's got wonderful illustrations and um and they they seem to have been on to idea that the projected geometry was crucial to Consciousness and mind um and but you know they also had a lot of strange other Notions that that made them seem like cranks you know um and then there was another um interest in just higher dimensional geometry that came from people like um uh PD ensky if you've heard of this of this this guy um he um he was he wrote book called the fourth dimension and his his view is that there must be part of our selves that we don't have access to in the in in the way in which we have access to three-dimensional objects which kind of and he thought that would be in the fourth dimension it's not clear that he thought that it was a projective geometrical structure but um it's still a sort of you know sort of in the ballpark um well just one note there is that uh ensky who studied under uh GFF there some deep esoteric threads and of course going much deeper Through Time alternate ideas about number geometry color and of course philosophy those are in many ways what define the uh less religious at least in the exoteric sense of these wisdom Traditions so it's very interesting to see the geometry spiritual connection return after who knows maybe never even a separation but at the very least in most people's mind they' think of it as shapes on a plane not about the conceptions of the lower level structure of reality or the higher level structures of reality like the shape of the universe Shape of Things shape of ourselves so very nice threads and David's a big fan of uh what's his name matila geeka isn't that right yeah poet mathematician and Romanian prince who wrote many book of about the connection between mathematic Aesthetics and natural form yes there are a lot of connection that were very inspiring to me my father used to show me his book when I was a kid and my grandfather was an art historian and he was also searching you know interpretation of art forms through mathematics so there is a long family tradition that led me here so I'm sorry but I will have to leave to another call it was very very great I can leave you and you can continue discuss and was very happy so uh we'll get in touch and and stay in touch and I'll see you great bye bye see you I'll um move us to the end but really uh D and Ken thanks so much it's like always just great to have the authors on here and to be hearing all this extra stuff so the other slides just to you know visualize them people could screen capm is just a few more quotes to get to that he hegelian dialectic which we actually got to talk about um we could have had a few more just general questions especially where does active inference fit in into this whole picture and that's in the cybernetics and how it's carried out but uh active inference live stream and whatnot that's always what we want to think about and then we just

had figure one figure two figure three and figure four so those are just things that we could have talked about maybe uh in the future or in 2021 we could have the Moon Illusion or a deep dive on one dimension or another because this was such a fascinating conversation so if uh Adam or Alex or Ken maybe each of you want to have just a uh last comment that would be great so let's um Alex if you want to go first then Adam and then Ken or pass no worries if not okay okay I'll pass by now thank you Alex um Adam any last thoughts um how's it gonna change what you're G to work on or think about or just um too many thoughts but one of the things I'm uh I'm doing a lot of thinking about like what neural systems uh might potentially contribute to different aspects of uh uh visual spatial awareness and geometric modeling and um one issue I'm wondering about um is the extent to which uh at least its visual nature it might always the conscious awareness might always be um more of of a of a 2d projection and that um things like threedimensional so and things like additional depth and additional Dimensions might involve um uh the ability to uh transition between different points of view in a coherent way in a way that would imply a 3D world but where at any given moment the visual aspect um is uh actually uh 2D and potentially without clear depth um these are some things I'm wondering about and I think it's not incompatible with projective geometric modeling it would just be um whether the um the engine you know how distributed is and like the granularity over which it's being achieved and so this like coupling between like a 2D projection screen and like you know a body map like adjusting what you ought to see or something like the posterior singulate like saying like okay here's where you're like your eyes and your head is so you know giving putting your perspective in there and then putting it seems like 40 geometry could be perfectly compatible with this and but I'm trying to figure out how that works that was a lot of rambling but thank you no great thanks for your participation and this figure two I think points out some of what you're talking about that's a projection onto the 2D there's an ambiguous projection on the left because of the perspective being ambiguous and then free energy is invoked as like a symmetry break that gives a perspective on which side of the box one is actually on given a prior that about how certain angles must be reflective of two-dimensional representations so yeah Ken any um thoughts on that and or just a last comments uh well just thank you for organizing this and I'm I'm sorry I couldn't make it for the whole thing um and it's really been a wonderful conversation and I'll just say the uh the next thing we're working on the Daniel and d and I is a paper trying to relate the projective geometrical model uh close or to show how it relates to Notions like strange strange and um pre-reflective self-consciousness this kind of more basic type of self-consciousness than uh

introspection or or attention and to show that you can you can make sense of that in terms of the projective geometrical structure that's the next thing on the agenda for us cool sounds like a fun research Direction so with all of this being said everyone thanks a lot for participating live and if you're watching for watching it too we will have the follow-up form link is in the calendar invite if you want to just provide us some thoughts um any feedback suggestions and questions are always welcome other than that just uh stay in communication really appreciate this great conversation and some awesome stuff so thanks again everyone for participating and watching and we will talk soon so see you later thank you